Blank / Junk Logic

As built on 2026-11-08. Code: `core-pipeline/modules/blank_junk/` (`method.py` the ladder, `stage.py` the short page and duplicates). Numbers: `resources/canon/blank_junk_context_canon.json`.

Each page gets one verdict: **Main**, **Blank**, or **Junk / subtype**. The subtypes are Cover Page, Cards, Letter/Fax, Record Request, Instructions and Others. There is no Junk / Invoice. Every rule is in the ladder below; nothing runs after it.

Two terms used throughout:

- **Electronic Signature** = kv_extract found a provider signature on the page. Stage 7 (kv_extract) writes `staging/extraction.json`, and from the same results one signature row per page (`imaging/<chart>_provider_signature.csv` and `provider_signature_results`). Blank / junk reads `signature_present = y`, and the signature's region and sentence, from that row. Words such as "signed" or "signature" in the text never count on their own.
- **Previous / next page** = the nearest page before / after that is **not blank** (no text, or no ink in stage 1). Blank pages in between are skipped.

Pass 1: on the Tesseract text, rules only

The first rule that matches decides.

#	Check	Verdict
1	No ink, stage 1 says blank, no text, or the word "blank" in < 30 characters	Blank
2	"Intentionally left blank" on a page of ≤ 25 words	Blank
3	A cover mark (Accept, Unaccept, Cover) on a page of < 10 words	Junk / Cover Page
4	"End of Report" on a page of < 10 words	Junk / Cover Page
5	< 15 characters	Junk / Others
—	Anything else	Main for now. Pass 2 decides

Pass 2: on the best Final OCR text

The first rule that matches decides.

#	Rule	Check	Verdict
1	Blank	The pass 1 rules 1–5 again	as in pass 1
2	Visual	A visual page under 20 words that is not a card	Main
3	Handwritten	More than 30% handwritten	Main
4	**Signature**	An Electronic Signature here, **and** none on the previous or next page, **and** the model's p(junk) < 0.95	Main
5	Lab result	One word from each group: **Laboratory / Lab Result(s) / Lab Report / Lab Test(s)** (not "Lab" alone) · **Reference Range** (Ref Range, Reference Interval) · **Value(s)**	Main
6	Clinical content	2+ ICD-10 codes (I11.0, E11.9…), or a Social History header	Main
7	Card	An ID or insurance card. The model's p(keep) ≥ 0.99 keeps it	Junk / Cards
8	Fax	"Fax" in a section header, and rule 4 did not keep the page	Junk / Letter/Fax
9	Contents page	"Index" / "Contents" / "Table of Contents" as a heading, plus 3+ lines with page numbers	Junk / Others
10	**Short page**	Content ends in the top 20% of the page. See below	see below
11	Classifier	The model says JUNK, but p(keep) ≥ 0.99	Main
		The model says Invoice or insurance ID, and it is a card (the card rule nearly matched, or the model's insurance-ID label)	Junk / Cards
		Otherwise Invoice, with a money amount **and** an **Invoice Date** or **Due Date** with a date	Main (page type Invoice / Invoice Page)
		Otherwise Invoice	Junk / Others
		The model says JUNK of another kind	Junk / its subtype
		The model says BLANK	Junk / Others
		Otherwise	Main

Rule 10: the short page

Rule 10 applies only to a page that got past rules 1–9. A page kept by the signature, lab or clinical rules is never junked here. The first check that matches decides.

#	Check	Verdict
a	**The signature carries on from the page before.** The previous page ends in its Electronic Signature (kv_extract places it in the footer, or its signature sentence is in that page's last paragraph). This page has the provider's name or credentials (kv_extract), the signer's name, credentials (MD, DO, NP…), or a date	Main
b	**Fewer than 5 distinct words** that are not on the previous or next page	Junk / Others
	…unless those words are only "Diagnosis" / "Visit Diagnosis"	Main
—	Otherwise	the classifier decides (rule 11)

An invoice page is never junked by the short page rule.

Duplicates (pass 2, after the verdict)

These checks compare only Main pages. The result is **Duplicate**, **Partial**, or nothing; Repeated Content is off.

- **Duplicate:** the two pages are about the same length (the shorter at least 90% of the longer), and either the shorter page is contained in the other, ≥ 98% of the words overlap, or fewer than 8 distinct words differ.
- **Partial:** 95–98% overlap, or fewer than 15 distinct words differ. Length is not checked, so a page inside a much longer one is a Partial at most.

The later page is always the duplicate. Two pages are never matched when:

- their dates of service, or any date written on them, differ;
- a signed page (rule 4) is apart from its look-alike (another Main page between them).

Pass 1 marks only exact copies up to 3 pages apart.

Continuity

Off for now: continuity no longer turns a junk page inside a Progress Note back into Main.